The Squiggly Line Conjecture: A Geometric Picture of Overparameterization That Training Cannot Use

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Abstract

The squiggly line conjecture is a geometric account of why larger networks learn faster: a network with more parameters is a squigglier random curve at initialisation, a squigglier curve passes nearer to the data, and so a larger network starts closer to a fit. We split it into three claims and test each. The geometric claim holds. For random-phase sines of fixed amplitude, the mean Euclidean distance from a random point to the nearest point of the curve falls by a factor of 30 as the frequency rises from 0.25 to 64. The other two claims fail. The distance that falls is not the one a regression loss charges: for a K-Lipschitz function the nearest-point distance lies between the vertical error divided by $\sqrt{1+K^2}$ and the vertical error, so a curve can be much nearer than its vertical error only where it is steep. The expected loss of a random function depends only on its one-point marginals, and for random-phase waves these do not depend on the frequency; in the same experiment the mean vertical error stays at 0.67. For random ReLU networks under He initialisation the expected initial squared error is the same at every width, and the curves of random networks do not get longer with width. We also prove a lower bound on the mean distance in terms of curve length, derive the constant for the sine, and identify what sets the saturation floor. The main statements are checked in Lean 4. The picture is correct as geometry and does not explain learning.

1 Introduction

Networks with far more parameters than training points are easier to train than small ones, and several accounts of this exist: a large random network contains good sparse subnetworks [Frankle and Carbin, 2019, Ramanujan et al., 2020]; a wide network stays close to its linearisation around the initial weights [Jacot et al., 2018, Chizat et al., 2019]; and test risk can fall again past the interpolation threshold [Belkin et al., 2019, Nakkiran et al., 2020]. This paper examines a simpler geometric picture, which we call the *squiggly line conjecture*, and reports that it does not work.

The picture is as follows. Scatter random points on a plane and draw two curves without looking at the points, one smooth and one squiggly. The squiggly curve passes closer to the points on average, because it visits more of the plane (Figure 1). A network computes a function, and the graph of that function is such a curve. The conjecture says that a network with more parameters is a squigglier curve at initialisation, that it therefore starts closer to its data, and that this is why it learns faster. This is a chain of three claims.

(G) Geometry. A squigglier curve has a smaller mean Euclidean distance from a data point to the nearest point of the curve.

(P) Parameters. More parameters give a squigglier random function.

(L) Learning. A smaller nearest-point distance means a smaller training loss at initialisation.

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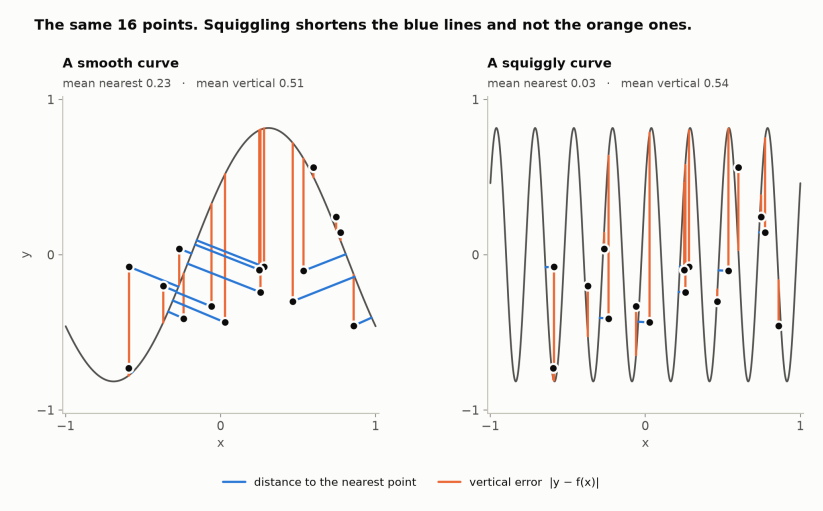


Figure 1: The two distances, on the same 16 points. Blue: distance to the nearest point of the curve. Orange: vertical error $|y - f(x)|$. Raising the frequency of the sine from 1 (left) to 8 (right) shortens the blue segments and leaves the orange ones as long as before.

Results. (G) is true, (P) is false for width and holds weakly for depth, and (L) is false.

- *Slope sandwich* (Theorem 1). For a K-Lipschitz function and any data point, $\text{vertical}/\sqrt{1+K^2} \le \text{nearest} \le \text{vertical}$. The curve can be much nearer than its vertical error only where it is steep.
- *Marginal lemma* (Theorem 2). The expected loss of a random function, for any loss of the residual $y - f(x)$, depends only on the distribution of $f(x)$ at each single x. For random-phase waves this distribution does not depend on the frequency (Corollary 3), and for ReLU networks under He initialisation the expected initial squared error does not depend on the width (Corollary 4).
- *Tube identity and length bound* (Theorem 5 and Proposition 7). The mean distance from a random point to a set is the mass outside the tube of radius r around the set, integrated over r. For a curve of length l and a uniform point in a planar region of area V it is at least $V/(4l + 2\sqrt{\pi V})$.
- *The sine* (Propositions 8 and 9). For a sine that spans the data, distance times frequency tends to $(2 - 8/\pi^2)/4 \approx 0.2974$. With a smaller amplitude the distance stops at a floor set by the amplitude, not by the number of points.
- *Experiments* (Section 3). With amplitude and marginals held fixed, the nearest-point distance falls from 0.613 to 0.021 while the vertical error stays between 0.665 and 0.669. Random ReLU networks have the predicted, width-independent initial squared error, and their curves are not longer at larger width.

Scope. This is a negative result about one intuitive picture. An earlier draft of this paper claimed empirical support for the conjecture; that support came from faulty measurement code, and the draft was withdrawn (Appendix B). The individual statements here are elementary, and we make no claim of priority for any of them; Section 5 lists the work we are aware of. Theorems 1, 2 and 5 and Corollary 3 are formalised in Lean 4 (Section 4). Code, the Lean project and the development notes are at `https://github.com/Ghost---Shadow/squiggly-line-conjecture`.

2 Setting and results

2.1 Two distances

Let (X, d) be a metric space of inputs and $f : X \to \mathbb{R}$ a function. A data point is a pair $(x, y) \in X \times \mathbb{R}$. We give $X \times \mathbb{R}$ the Euclidean product metric $\rho\big((x, y), (x', y')\big) = \sqrt{d(x, x')^2 + (y - y')^2}$ and write

$\Gamma_f = \{(x, f(x)) : x \in X\}$ for the graph of f, the "line" of the conjecture. Two quantities compare a data point with the line:

$$v_f(x,y) = |y - f(x)| \qquad \text{(vertical error)},$$
$$\delta_f(x,y) = \inf_{x' \in X} \rho\big((x,y),(x',f(x'))\big) \qquad \text{(nearest-point distance)}.$$

Regression losses are functions of the residual $y - f(x)$, so they see v_f. The conjecture is stated with δ_f.

A *random function* F is a probability space (Θ, π) of parameters together with functions $f_\theta : X \to \mathbb{R}$ such that $\theta \mapsto f_\theta(x)$ is measurable for every x. Its *one-point marginal* at x is the law ν_x^F of $f_\theta(x)$ under π. For a cost c and a data distribution μ on $X \times \mathbb{R}$, the expected cost at initialisation is $\mathbb{E}_{(x,y)\sim\mu}\, \mathbb{E}_{\theta\sim\pi}\, c(f_\theta; x, y)$.[1] For a family of random functions indexed by a size m, "a bigger model starts closer" means that this expectation is strictly decreasing in m. Claim (G) uses the cost δ and claim (L) uses costs of the form $\ell(y - f(x))$. We make "squiggly" concrete by the frequency of a wave of fixed amplitude and, for networks, by arc length and the number of turns.

2.2 The slope sandwich

Theorem 1 (Slope sandwich)**.** *If f is K-Lipschitz, then for every data point (x, y)*

$$\frac{v_f(x,y)}{\sqrt{1+K^2}} \;\le\; \delta_f(x,y) \;\le\; v_f(x,y).$$

Proof. The point $(x, f(x))$ lies on Γ_f at distance $|y - f(x)|$ from (x, y), which gives the upper bound. For the lower bound take any $x' \in X$ and put $a = d(x, x')$ and $b = |y - f(x')|$. By the triangle inequality and the Lipschitz bound, $|y - f(x)| \le |y - f(x')| + |f(x') - f(x)| \le b + Ka$. By the Cauchy–Schwarz inequality, $b + Ka \le \sqrt{1+K^2}\,\sqrt{a^2+b^2} = \sqrt{1+K^2}\;\rho\big((x,y),(x',f(x'))\big)$. Taking the infimum over x' gives $v_f(x,y) \le \sqrt{1+K^2}\,\delta_f(x,y)$. □

Both bounds are attained: for $X = \mathbb{R}$ and $f(x) = Kx$, the point $(0, y)$ has $\delta_f = |y|/\sqrt{1+K^2}$, and for a constant f the two distances coincide. Taking expectations, the same two inequalities hold between the mean distances of any random K-Lipschitz function and any data distribution.

The theorem says what a small nearest-point distance means. If $\delta_f(x,y) < \eta$, there is an input x' with $d(x,x') < \eta$ and $|y - f(x')| < \eta$: a perturbation of the *input* smaller than η brings the output within η of the label. When δ_f is small and v_f is not, the function is steep near x: its Lipschitz constant is at least $\sqrt{(v_f/\delta_f)^2 - 1}$. That describes a function that is sensitive to its input near the data, not one that fits the data.

2.3 The marginal lemma

Theorem 2 (Marginal lemma)**.** *Let F be a random function and $\ell : \mathbb{R} \to [0, \infty)$ a measurable loss. For every data point (x, y),*

$$\mathbb{E}_\theta\, \ell\big(y - f_\theta(x)\big) = \int_{\mathbb{R}} \ell(y - t)\, \nu_x^F(dt).$$

Consequently, if two random functions F and G have $\nu_x^F = \nu_x^G$ for every x, their expected losses are equal for every such ℓ and every data distribution μ.

Proof. By definition ν_x^F is the image of π under $\theta \mapsto f_\theta(x)$. The displayed identity is the change-of-variables formula for an image measure, applied to the measurable function $t \mapsto \ell(y - t)$. Integrating both sides against μ gives the second statement. □

The proof is one line, and the content is in what is absent: the joint law of $\big(f_\theta(x), f_\theta(x')\big)$ at two inputs does not enter. How f moves between inputs is what "squiggly" means, so the expected initial

[1] We assume the inner expectation is a measurable function of (x, y). This holds if $(\theta, x) \mapsto f_\theta(x)$ is jointly measurable, and always when μ is the empirical measure of a finite data set.

loss cannot register squiggliness. The lemma is about the expectation over θ only. The variance of the initial loss, the probability of a low initial loss, and everything that happens during training do depend on joint laws.

Corollary 3 (Random-phase waves)**.** *Let $g : \mathbb{R} \to \mathbb{R}$ be measurable and T-periodic, let φ be uniform on $[0, T)$, and for a frequency $w \in \mathbb{R}$ let $f_{w,\varphi}(x) = g(wx + \varphi)$ on $X = \mathbb{R}$. Then the one-point marginal of $f_{w,\cdot}$ at every x is the law of $g(\varphi)$, for every w. Hence the expected loss $\mathbb{E}_{(x,y)\sim\mu}\mathbb{E}_\varphi\, \ell\big(y - f_{w,\varphi}(x)\big)$ is the same for every w, every measurable $\ell \geq 0$ and every μ. No sequence of frequencies makes it decrease.*

Proof. The uniform distribution on the circle $\mathbb{R}/T\mathbb{Z}$ is invariant under translation, so $wx+\varphi \bmod T$ is uniform on $[0, T)$ for every fixed w and x, and $g(wx+\varphi)$ has the law of $g(\varphi)$. Apply Theorem 2. □

For the curves of Section 3, $g = p\sin$ with peak $p \leq 1$, and data with y uniform on $[-1, 1]$ independent of x. Since $\mathbb{E}_y|y - s| = (1 + s^2)/2$ for $|s| \leq 1$, the expected vertical error is $(1 + p^2/2)/2$ at every frequency.

Corollary 4 (ReLU networks under He initialisation)**.** *Let $f_\theta(x) = u^\top \sigma\big(W^L \sigma(\cdots \sigma(W^1 x))\big)$ for $x \in \mathbb{R}^d$, with σ the ReLU, no biases, hidden widths $n_1, \ldots, n_L$, $n_0 = d$, and independent weights $W^k_{ij} \sim N(0, 2/n_{k-1})$ and $u_i \sim N(0, 2/n_L)$. Then for every x and y,*

$$\mathbb{E} f_\theta(x) = 0, \qquad \mathbb{E} f_\theta(x)^2 = \frac{2\|x\|^2}{d}, \qquad \mathbb{E}\big(y - f_\theta(x)\big)^2 = y^2 + \frac{2\|x\|^2}{d},$$

whatever the depth and the widths. If all weights are instead $N(0, 1)$, the last quantity is $y^2 + \|x\|^2 \prod_{k=1}^{L}(n_k/2)$.

The proof is a layer-by-layer recursion for $\mathbb{E}\|h^k\|^2$ and is in Appendix A. It uses only that the weights are independent, symmetric and of the stated variance; preserving the second moment across layers is the design criterion of this initialisation [He et al., 2015]. In the terms of Theorem 2, the squared loss reads only the first two moments of ν_x, and these do not depend on the width. The full law ν_x does depend on the width, so for other losses the corollary gives no exact statement.

2.4 The tube identity and a bound from length

For a non-empty set S in a metric space, write $N_r(S) = \{q : \operatorname{dist}(q, S) \leq r\}$ for the closed tube of radius r around S.

Theorem 5 (Tube identity)**.** *Let S be a non-empty subset of a metric space and P a random point of that space. Then*

$$\mathbb{E}\, \operatorname{dist}(P, S) = \int_0^\infty \mathbb{P}\big(\operatorname{dist}(P, S) > r\big)\, dr = \int_0^\infty \mathbb{P}\big(P \notin N_r(S)\big)\, dr,$$

where both sides may be infinite.

Proof. The function $q \mapsto \operatorname{dist}(q, S)$ is 1-Lipschitz, hence measurable, and non-negative. For a non-negative random variable Z we have $Z = \int_0^\infty \mathbf{1}\{Z > r\}\, dr$, and Tonelli's theorem gives $\mathbb{E}Z = \int_0^\infty \mathbb{P}(Z > r)\, dr$. This is the layer-cake formula [Lieb and Loss, 2001, Section 1.13]. Take $Z = \operatorname{dist}(P, S)$. □

The identity holds for any set in any metric space. It reduces the mean distance to the volume of tubes, which for curves in the plane is controlled by length.

Lemma 6. *Let $S \subset \mathbb{R}^2$ be the image of a rectifiable curve of length l. Then the area of $N_r(S)$ is at most $2lr + \pi r^2$ for every $r \geq 0$.*

The proof, by polygonal approximation, is in Appendix A.

Proposition 7 (Length bound)**.** *Let $\Omega \subset \mathbb{R}^2$ be measurable with area $V \in (0, \infty)$, let P be uniform on Ω, and let S be the image of a rectifiable curve of length l. Then*

$$\mathbb{E}\, \operatorname{dist}(P, S) \;\geq\; \frac{V}{4l + 2\sqrt{\pi V}}.$$

Proof. Put $r_0 = V/(2l + \sqrt{\pi V})$. For $0 \le r \le r_0$ we have $\pi r \le \pi V/\sqrt{\pi V} = \sqrt{\pi V}$, so by Lemma 6 the area of $N_r(S)$ is at most $(2l + \sqrt{\pi V})\, r = V r/r_0$. Therefore $\mathbb{P}(P \notin N_r(S)) \ge 1 - r/r_0$ for $r \le r_0$, and by Theorem 5 $\mathbb{E}\ \mathrm{dist}(P, S) \ge \int_0^{r_0} (1 - r/r_0)\, dr = r_0/2$. □

To leading order in l the bound is $V/(4l)$. We draw three consequences. Length is necessary: a mean distance of at most ε requires $l \ge V/(4\varepsilon) - \sqrt{\pi V}/2$. Length is not sufficient: a part of the curve further than r from Ω adds nothing to $N_r(S) \cap \Omega$, so length outside the data region is wasted (Section 3.2). And the bound is nearly attained only if the tube around the curve stays inside Ω and overlaps itself little, that is, if the length is spread evenly over the region. The proposition is for curves in the plane; for inputs of dimension above one the graph is a surface, and its area takes the role of length.

2.5 The sine family

The experiments use $f_{w,\varphi}(x) = p \sin(w\pi x + \varphi)$ on $x \in [-1, 1]$ with φ uniform on $[0, 2\pi)$, and a point P uniform on $\Omega = [-1, 1]^2$, so $V = 4$. The curve makes w full periods across the box and is $p\, w\pi$-Lipschitz. Write $D_p(w)$ for the mean of δ over P and φ, and $l(w)$ for the arc length.

Proposition 8 (The sine that spans the data; leading order)**.** *For $p = 1$, as $w \to \infty$,*

$$\frac{l(w)}{w} \to 4, \qquad w\, D_1(w) \to \frac{2 - 8/\pi^2}{4} \approx 0.2974, \qquad D_1(w)\, l(w) \to 2 - \frac{8}{\pi^2} \approx 1.189.$$

The first limit is proved in Appendix A. The second is a leading-order derivation. At height y the curve crosses on two interleaved lattices of period $2/w$, with gaps $(1/w)(1 \mp 2 \arcsin(y)/\pi)$. At high frequency the strokes are nearly vertical, so the nearest point of the curve is, to leading order, the nearest crossing at the same height. Averaging that distance over x and then over y, with $\int_0^1 \arcsin(y)^2\, dy = \pi^2/4 - 2$, gives the constant. The step from horizontal to Euclidean distance is argued in outline in Appendix A, without all details, and the constant is checked numerically in Section 3.3. The limit 1.189 exceeds the value $V/4 = 1$ of Proposition 7 because the strokes of a sine are not evenly spaced.

Proposition 9 (Band-edge floor)**.** *For $0 < p \le 1$ and $w \ge 1$,*

$$\frac{(1-p)^2}{2} \ \le\ D_p(w)\ \le\ \frac{(1-p)^2}{2} + \frac{2}{w}.$$

The proof is in Appendix A. The curve stays in the band $|y| \le p$, so a point at height $|y| > p$ is never nearer than $|y| - p$, and $\mathbb{E} \max(0, |y| - p) = (1-p)^2/2$. Raising the frequency removes the rest at rate $O(1/w)$. The floor depends on the amplitude relative to the range of the data. It does not depend on the number of data points: $D_p(w)$ is the mean of a per-point quantity, so its expectation is the same for a sample of any size.

3 Experiments

All experiments are small numerical computations with our own scripts (Appendix C lists them). Unless stated otherwise, curves are $f(x) = A\sqrt{2}\, \sin(w\pi x + \varphi)$ on $[-1, 1]$ and data points are uniform on $[-1, 1]^2$, with y independent of x. The default amplitude is $A = 0.577$, the standard deviation of the labels, so the peak is $p = A\sqrt{2} = 0.816$. Each curve is sampled as a polyline with consecutive samples at most 0.002 apart along the curve, and the distance from a point to the polyline is computed exactly.

3.1 Frequency with everything else held fixed

Earlier tests of the conjecture varied polynomial degree or network width, which changes the output amplitude together with the squiggliness. Here the frequency w is the only variable: by Corollary 3 the distribution of $f(x)$ at every x is identical in every row of Table 1. We use 3000 trials of 30 points each and 100 stratified random phases per frequency.

Table 1: Frequency isolation. Mean nearest-point distance and mean vertical error against frequency w, with the two lower bounds on the nearest-point distance: the mean vertical error divided by $\sqrt{1+K^2}$ with $K = p\,w\pi$ (Theorem 1), and $4/(4l + 4\sqrt{\pi})$ (Proposition 7 with $V = 4$).

w	arc length l	nearest-point	vertical	bound, Theorem 1	bound, Proposition 7
0.25	2.19	0.6129	0.6672	0.5617	0.2524
0.5	2.66	0.4890	0.6664	0.4099	0.2256
1	3.96	0.3095	0.6649	0.2416	0.1744
2	6.96	0.1717	0.6651	0.1273	0.1145
4	13.32	0.0928	0.6663	0.0647	0.0663
8	26.26	0.0529	0.6678	0.0325	0.0357
16	52.31	0.0338	0.6665	0.0162	0.0185
32	104.50	0.0249	0.6657	0.0081	0.0094
64	208.92	0.0207	0.6688	0.0041	0.0047

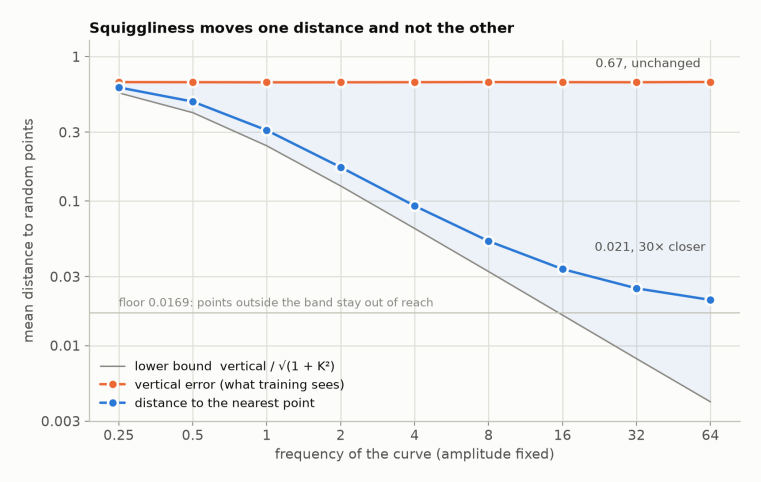


Figure 2: The data of Table 1. The nearest-point distance (blue) falls with frequency and stays above the lower bound of Theorem 1 (grey line). The vertical error (orange) does not move. The horizontal line is the floor $(1-p)^2/2 = 0.0169$ of Proposition 9.

The nearest-point distance falls from 0.6129 to 0.0207, a factor of 29.6. Claim (G) holds, and not through amplitude. The vertical error lies between 0.6649 and 0.6688 in every row, with no trend; the value predicted after Corollary 3 is $(1 + p^2/2)/2 = 0.6665$. The flat column is Corollary 3 observed, and the differences between rows are sampling noise. Both lower bounds hold in every row. At $w = 64$ the function has Lipschitz constant 164, and this steepness is what Theorem 1 requires for a nearest-point distance 32 times smaller than the vertical error. From $w = 16$ the gap to both bounds widens, because the distance approaches the floor of Proposition 9 (Figure 2).

3.2 A straight line of the same length

If length alone produced the effect, a straight segment of the same arc length should score like the sine. We take $y = mx + b$ on $x \in [-1, 1]$ with $m = \sqrt{(l/2)^2 - 1}$, so that its length is l, and b uniform on $[-1, 1]$.

The segment stays at 0.50 while the sine reaches 0.021 (Table 2). This comparison needs a caveat. A segment of length 209 over $x \in [-1, 1]$ has slope 104, and about 99% of it lies above or below the data. The part inside the box has length about 2 in all three rows, and a near-vertical segment of length 2 through the middle of the box has mean distance $\mathbb{E}|x| = 0.5$. The table therefore shows that length outside the data region is worth nothing. It does not show that length is irrelevant: by Proposition 7, length is necessary, and the sine achieves its small distance because its length is inside the box and spread across it.

Table 2: Mean nearest-point distance for the sine and for a straight segment of the same length. The last two columns give the slope of the segment and the share of its length that lies outside the data box $[-1, 1]^2$, averaged over b.

w	arc length	sine	straight segment	slope m	share outside the box
4	13.32	0.0928	0.5022	6.6	85%
16	52.31	0.0338	0.5007	26.1	96%
64	208.92	0.0207	0.5006	104.5	99%

Table 3: Knee frequency w^* against the number of points n. The prediction under test was slope 1 of $\log w^*$ on $\log n$. Confidence intervals are from a bootstrap over trials.

criterion	$n = 10$	30	100	300	1000	slope	95% interval
two-segment fit	21.6	21.6	21.6	21.6	21.6	-0.000	$[-0.004, 0.004]$
doubling gains $< 10\%$	60.1	61.7	61.4	61.0	59.4	-0.003	$[-0.014, 0.008]$

3.3 What sets the floor

An earlier version of this work predicted that the distance stops falling once the wavelength $2/w$ drops below the mean spacing $2/n$ of n data points, so that the knee of the curve $D(w)$ would sit at $w^* \propto n$. We swept $n \in \{10, 30, 100, 300, 1000\}$ and 25 frequencies in $[1, 1000]$ with 30 000 points per n, and located the knee by two criteria fixed in advance: the breakpoint of a continuous two-segment fit of $\log D$ on $\log w$, and the smallest w at which doubling w lowers D by less than 10%.

The slope is 0, not 1 (Table 3), and the five curves $D(w)$ agree to within 3.1% at every frequency. No experiment was needed for this: the quantity is a mean of per-point distances, so its expectation cannot depend on n. What sets the knee is the band edge of Proposition 9. With $p = 0.816$ the floor is $D_\infty = (1 - p)^2/2 = 0.0169$, and a fit of $D - D_\infty = c\, w^{-\beta}$ for $w \geq 20$ gives $\beta = 0.990$ and $c = 0.248$. For $D = D_\infty + c/w$ the 10% criterion triggers at $c(1 - 2r)/(2rD_\infty) = 58.6$ with $r = 0.1$, against 59 to 62 measured.

As a control we set $A\sqrt{2} = 1$, so that the band covers the data and the floor is zero. The saturation disappears (Figure 3): the distance follows $0.301\, w^{-1.002}$ up to $w = 1000$, where it is 0.00030, and the 10% criterion never triggers.[2] Proposition 8 predicts $w\, D_1(w) \to 0.2974$. The repository reports 0.2965 at $w = 256$; a separate run for this paper, with $40\,000$ points and 8 phases, gave 0.298 at $w = 64$ and at $w = 256$, and $D\, l = 1.192$ against the predicted 1.189.

3.4 Random networks

Initial loss against width. We draw 200 points with x uniform on $[-1, 1]^{10}$ and y uniform on $[-1, 1]$, and for each width 200 untrained ReLU networks with two hidden layers and no biases, as in Corollary 4. The prediction of Corollary 4, evaluated on the sample, is the mean of $y^2 + 2\|x\|^2/d$, which is 1.03.

A wider network does not start closer (Table 4; Figure 4 in Appendix C). The measured values scatter around the prediction; the largest deviation, at width 256, is 1.9 standard errors. With unscaled weights a wider network starts much further away, because its output grows with width. That growth is an amplitude effect, and it is the confound that the design of Section 3.1 removes.

Curve length against width and depth. For claim (P) we look at the curves themselves, for a one-dimensional input. Each network has He-uniform weights and hidden biases uniform on $[-0.5, 0.5]$, and its output is shifted and scaled so that its range over $[-1, 1]$ is exactly $[-1, 1]$. We measure arc length, the number of turns (sign changes of the slope), and the mean nearest-point distance to 200 fixed random points, over 300 networks per size.

[2] The two-segment fit still reports a breakpoint, at 1.8, because it always returns one.

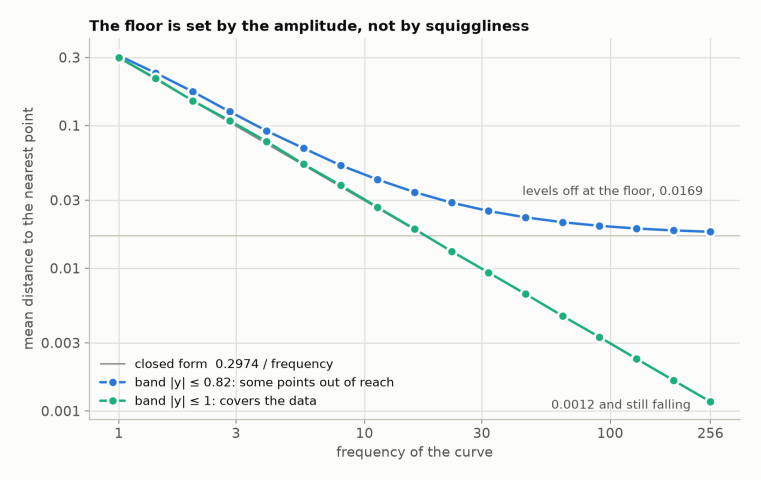


Figure 3: The floor comes from the amplitude. Blue: default amplitude, band $|y| \leq 0.82$; the distance levels off at 0.0169. Green: band $|y| \leq 1$; no floor, and the distance follows the constant of Proposition 8 (grey line, under the green one).

Table 4: Mean squared error before training, 200 networks per width ($\pm$ one standard error). Predictions are from Corollary 4.

	He initialisation		$N(0,1)$ weights	
width	measured	predicted	measured	predicted
4	1.07 ± 0.12	1.03	15	14
16	1.02 ± 0.05	1.03	220	219
64	1.07 ± 0.04	1.03	3.45×10^3	3.51×10^3
256	1.10 ± 0.04	1.03	6.02×10^4	5.61×10^4
1024	1.03 ± 0.04	1.03	8.23×10^5	8.97×10^5

Width adds turns and no length (Table 5 in Appendix C). From width 2 to 512, with two hidden layers, the mean number of turns rises from 0.5 to 15, the arc length stays between 3.4 and 3.6, and the nearest-point distance is flat from width 16, between 0.42 and 0.43. Depth adds some length: from 1 to 16 layers at width 64 the arc length rises from 3.4 to 5.1 and the distance falls from 0.44 to 0.34. For comparison, the same code gives length 4.61 and distance 0.351 for a sine with one period across the box, and 64.07 and 0.021 for a sine with 16 periods. Even at depth 16 the random curves resemble the first and not the second. So claim (P) fails for width and holds weakly for depth.

This measurement has caveats. The input is one-dimensional. Rescaling the output controls the amplitude but does not hold the marginals of $f(x)$ fixed, unlike Section 3.1. Only the nearest-point distance is measured, not the vertical error. The distances use a fixed grid of 4001 samples, which overestimates the distance by about 7% for the sine with 16 periods (0.0209 against 0.0196 with exact projection). Finally, the rescaling and the biases mean that the depth trend is not directly comparable with the theorem of Hanin et al. [2022] discussed in Section 5.

4 Formalisation

The repository contains a Lean 4 project [de Moura and Ullrich, 2021] on Mathlib [The mathlib Community, 2020] that states the conjecture and proves the results that refute claim (L). It contains no `sorry`, and the repository reports that every theorem depends only on Lean's three standard axioms. The files were checked with Lean 4.36.0-rc1 against a pinned Mathlib commit.

What is stated. A randomly initialised model is a structure `RandomFunction` with a parameter type, a probability measure on it and an evaluation map. One definition, `StartsCloser`, says that

the expected cost is strictly decreasing in the size of the model. The conjecture is `StartsCloser` for the nearest-point distance, and the learning claim is `StartsCloser` for a loss of the residual. Neither is proved.

What is proved. Theorem 1 is `nearestDist_le_verticalDist` and `verticalDist_le_of_lipschitz`, for inputs in any seminormed group. Theorem 5 is `integral_infDist_eq_integral_uncovered`, for any non-empty set in a pseudo-metric space, under the hypothesis that the distance is integrable. Theorem 2 is `expected_residualLoss_congr`, and the first half of Corollary 3 is `marginal_randomPhaseWave_eq`, proved from the translation invariance of Haar measure on the circle. They combine into `not_learningClaim_sineWave`: for sines of any amplitude, any sequence of frequencies, any measurable loss $\ell : \mathbb{R} \to \mathbb{R}$ and any measure μ on data points, the learning claim fails. In Lean the expected cost is a Bochner integral, which is zero by convention when the integrand is not integrable; the theorems carry no integrability hypothesis on the loss and hold under that convention.

What is not formalised. Claim (G) itself, that frequency lowers the nearest-point distance, is measured and not proved. Lemma 6 and Propositions 7 to 9 and Corollary 4 have the pen-and-paper arguments given here and no formal proof.

5 Related work

We relate each claim to work we are aware of, and describe that work only as far as its abstract does. Our reading of how each bears on the conjecture is our own.

Length of random network curves (claim P). Raghu et al. [2017] propose trajectory length, the length of the image of a one-dimensional input path, as a measure of expressivity, and report that it grows exponentially with depth. Hanin et al. [2022] prove that for ReLU networks with standard random initialisation the expected length distortion does not grow with depth. Hanin and Rolnick [2019] show that the average number of activation patterns of a ReLU network at initialisation is bounded in terms of the number of neurons and the input dimension. These results bear on (P), and our length measurements in Section 3.4 agree with the second in spirit though not in setting. By Proposition 7, a curve that does not get longer cannot get much closer in the nearest-point sense either.

Squiggliness after training. Dherin et al. [2021] argue that stochastic gradient descent implicitly favours functions of low geometric complexity, which for one-dimensional regression is the arc length. Bubeck and Sellke [2021] show that, for broad classes of models and data, fitting the data smoothly requires d times more parameters than merely interpolating it, where d is the input dimension; in that account parameters buy smoothness, the opposite of (P). Humayun et al. [2024] report that the linear regions of a network migrate away from the training samples late in training. Hosseini and Fedorenko [2023] find that trained language models straighten the trajectory of a sentence across layers, and that untrained models do not. None of these supports the idea that a useful network is a squiggly one near its data.

Accounts in which size helps before or without training. Frankle and Carbin [2019] find sparse subnetworks that train in isolation to the accuracy of the full network. Ramanujan et al. [2020] show that a random network with fixed weights contains subnetworks that perform well without weight training, and that these improve as the random network gets wider and deeper. Larsen et al. [2022] find that training within a random subspace succeeds once its dimension passes a threshold that depends on the target loss and on the initial loss. Chiang et al. [2023] report that a guess-and-check optimiser, which samples random parameter vectors until one fits the training set, reaches test accuracy comparable to gradient descent in low-sample regimes. These are statements about subnetworks or about weight space. They do not concern the nearest-point distance of the full random function, and Corollary 4 is compatible with all of them. Martinelli et al. [2026] argue that the lottery-ticket analogy does not explain overparameterization and propose instead that width adds dimensions along which optimisation escapes bad minima.

Wide networks. In the lazy regime a model behaves like its linearisation around the initial weights [Chizat et al., 2019], with the neural tangent kernel governing training [Jacot et al., 2018], and wide random networks converge to Gaussian processes [Matthews et al., 2018]. This is the setting in which "starts closer" might be restated in weight space; we have not tested that reading (Section 7). The second-moment calculation behind Corollary 4 follows He et al. [2015].

6 Limitations

- *One-dimensional inputs.* The geometry experiments, Proposition 7 and Propositions 8 and 9 are for curves in the plane. Theorems 1, 2 and 5 hold in general, and Table 4 uses ten inputs, but claim (G) has not been measured for surfaces.
- *The sine family is degenerate.* For fixed w the functions $p\sin(w\pi x + \varphi)$ span a two-dimensional space and their covariance kernel $\frac{p^2}{2}\cos\big(w\pi(x - x')\big)$ has rank 2. No member can fit more than two generic points at once, at any frequency. The mean per-point distance cannot see this, because it scores each point alone.
- *Nothing to learn.* The labels are independent of the inputs, so no function predicts them. The experiments concern the loss at initialisation only and say nothing about learning speed or generalisation.
- *Expectation only.* Theorem 2 and Corollary 4 are about the mean initial loss. Whether the probability of a low initial loss changes with width is not addressed.
- *Own scripts.* The experiments are small computations written for this question, not an established benchmark, and the code has needed correction more than once (Appendix B).
- *Proof status.* The constant in Proposition 8 rests on a leading-order argument and a numerical check. Width enters the formal development nowhere.

7 Discussion

The conjecture joined two facts that are each true. A squigglier curve is nearer to random points, and larger networks learn faster. The link between them fails at both ends. Width does not make a random network's curve longer, and the nearness that squiggliness buys is nearness to points displaced in the input direction, which a regression loss does not reward. By Theorem 1 that nearness is paid for in slope.

Where "starts closer" might still hold. The claim could be moved from function space to weight space: a larger network may need to travel a shorter relative distance from its initial weights to a fit. The repository's notes predict, from the lazy-training results cited in Section 5, that it does, in which case an experiment would be a replication. We have not run it.

An untested proposal: joint fit. The nearest-point distance fails as a measure of fit partly because it is a mean over points. A cost that sees all points at once is the quadratic form $y^\top K^{-1} y$, where $y \in \mathbb{R}^n$ collects the labels and K is the positive-definite covariance matrix of the random function at the n inputs. It is the data-fit term of the Gaussian-process marginal likelihood [Rasmussen and Williams, 2006]. If the labels are uncorrelated with zero mean and unit variance, its mean is $\operatorname{tr}(K^{-1})$, and

$$\operatorname{tr}(K^{-1}) \;\geq\; \frac{n^2}{\operatorname{tr}(K)},$$

with equality if and only if K is a multiple of the identity. This is the Cauchy–Schwarz inequality $n^2 = \big(\sum_i \lambda_i^{1/2}\lambda_i^{-1/2}\big)^2 \leq \sum_i \lambda_i \sum_i \lambda_i^{-1}$ for the eigenvalues λ_i of K. At fixed output variance, $\operatorname{tr}(K)$ is fixed, so for random labels the cost is smallest for the prior whose values at different inputs are uncorrelated, which is the squiggliest one. For labels that vary smoothly with the input we expect the order to reverse. We state this as a direction only. Beyond the inequality above it is untested here: we have not measured this cost for networks, and we do not know whether it predicts anything about training.

Conclusion. The squiggly line picture is correct as geometry and does not explain why overparameterized networks learn. The distance it uses is not the one that training measures.

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A Proofs

A.1 Second moments under He initialisation

We prove Corollary 4. Put $h^0 = x$ and $h^k = \sigma(W^k h^{k-1})$. Given h^{k-1}, the coordinates of $W^k h^{k-1}$ are $N(0, s_k^2 \|h^{k-1}\|^2)$, where s_k^2 is the weight variance of layer k. For a centred Gaussian z, symmetry gives $\mathbb{E}\,\sigma(z)^2 = \frac{1}{2}\mathbb{E}z^2$. Hence $\mathbb{E}\big[\|h^k\|^2 \mid h^{k-1}\big] = \frac{1}{2} n_k s_k^2 \|h^{k-1}\|^2$. With $s_k^2 = 2/n_{k-1}$ this reads $\mathbb{E}\|h^k\|^2/n_k = \mathbb{E}\|h^{k-1}\|^2/n_{k-1}$, so $\mathbb{E}\|h^L\|^2/n_L = \|x\|^2/d$. Given h^L, the output is $N(0, (2/n_L)\|h^L\|^2)$, so $\mathbb{E}f_\theta(x) = 0$ and $\mathbb{E}f_\theta(x)^2 = 2\,\mathbb{E}\|h^L\|^2/n_L = 2\|x\|^2/d$. The third identity follows by expanding the square. With $s_k^2 = 1$ the recursion is $\mathbb{E}\|h^k\|^2 = \frac{1}{2} n_k \mathbb{E}\|h^{k-1}\|^2$ and $\mathbb{E}f_\theta(x)^2 = \mathbb{E}\|h^L\|^2$. □

A.2 Area of a tube around a curve

We prove Lemma 6.

Polygonal paths. Let Q be a polygonal path with vertices $q_0, \dots, q_k$ and segments of lengths $s_1, \dots, s_k$. For a segment $[a, b]$, the tube $N_r([a, b])$ is a rectangle of sides $|b - a|$ and $2r$ with a half-disc of radius r at each end. Its area is $2r|b - a| + \pi r^2$, and it contains the closed disc $\bar{B}_r(a)$, so $N_r([a, b]) \setminus \bar{B}_r(a)$ has area $2r|b - a|$. Let Q_j be the path up to q_j. Then $N_r(Q_0) = \bar{B}_r(q_0)$ has area πr^2, and $N_r(Q_j) = N_r(Q_{j-1}) \cup N_r([q_{j-1}, q_j])$. Since $\bar{B}_r(q_{j-1}) \subset N_r(Q_{j-1})$, passing from Q_{j-1} to Q_j adds at most the area of $N_r([q_{j-1}, q_j]) \setminus \bar{B}_r(q_{j-1})$, which is $2r s_j$. By induction the area of $N_r(Q)$ is at most $\pi r^2 + 2r \sum_j s_j$.

Rectifiable curves. Let $\gamma : [0, 1] \to \mathbb{R}^2$ be continuous with length $l < \infty$ and image S. Fix $\varepsilon > 0$. By uniform continuity there is a partition $0 = t_0 < \cdots < t_k = 1$ such that each arc $\gamma([t_{j-1}, t_j])$ has diameter at most ε. Let Q be the polygonal path through the points $\gamma(t_j)$; its length is at most l by the definition of length. Every point of S is within ε of a vertex of Q, and S is compact, so $N_r(S) \subset N_{r+\varepsilon}(Q)$. By the first step the area of $N_r(S)$ is at most $2l(r + \varepsilon) + \pi(r + \varepsilon)^2$. Let $\varepsilon \to 0$. □

A.3 Arc length of the sine

For $f(x) = \sin(w\pi x + \varphi)$ on $[-1, 1]$ the arc length is $l(w) = \int_{-1}^{1} \sqrt{1 + w^2\pi^2 \cos^2(w\pi x + \varphi)}\, dx$. Since $c \le \sqrt{1 + c^2} \le 1 + c$ for $c \ge 0$,

$$I \le l(w) \le 2 + I, \qquad I = w\pi \int_{-1}^{1} |\cos(w\pi x + \varphi)|\, dx = \int_{\varphi - w\pi}^{\varphi + w\pi} |\cos u|\, du.$$

The function $|\cos|$ has period π and integral 2 over a period. An interval of length $2\pi w$ contains $\lfloor 2w \rfloor$ full periods and a remainder shorter than one period, so $4w - 2 \le I \le 4w + 2$. Hence $4w - 2 \le l(w) \le 4w + 4$ and $l(w)/w \to 4$. With peak p the same argument gives $l(w)/w \to 4p$, in agreement with Table 1 ($4 \times 0.816 \times 64 = 208.9$).

A.4 The constant for the sine

We derive the second limit of Proposition 8.

Take $p = 1$ and fix a phase φ and a height $y_0 \in (-1, 1)$. Put $a = \arcsin(y_0) \in (-\pi/2, \pi/2)$.

Crossings. The curve is at height y_0 where $w\pi x + \varphi$ equals a or $\pi - a$ modulo 2π. These are two lattices in x, each of period $2/w$, and consecutive crossings are separated alternately by

$$g_1 = \frac{1}{w}\Big(1 - \frac{2a}{\pi}\Big), \qquad g_2 = \frac{1}{w}\Big(1 + \frac{2a}{\pi}\Big), \qquad g_1 + g_2 = \frac{2}{w}.$$

Horizontal distance. Let $h(x_0, y_0)$ be the distance from x_0 to the nearest crossing at height y_0. A point x_0 falls in a gap of length g with probability proportional to g, and its mean distance to the nearer end is then $g/4$. Averaging over x_0 across whole periods,

$$\frac{g_1^2 + g_2^2}{4(g_1 + g_2)} = \frac{1}{4w}\Big(1 + \frac{4a^2}{\pi^2}\Big).$$

The interval $[-1,1]$ contains whole periods and end pieces of total length at most $4/w$, on which $h \le 2/w$ for $w \ge 1$. So the mean of h over x_0 uniform on $[-1,1]$ differs from the display by $O(1/w^2)$.

Average over heights. For y_0 uniform on $[-1,1]$, with $y = \sin t$,

$$\mathbb{E}\,\arcsin(y_0)^2 = \int_0^1 \arcsin(y)^2\,dy = \int_0^{\pi/2} t^2 \cos t\,dt = \Big[t^2\sin t + 2t\cos t - 2\sin t\Big]_0^{\pi/2} = \frac{\pi^2}{4} - 2.$$

Therefore $w\,\mathbb{E}h \to \frac{1}{4}\big(1 + \frac{4}{\pi^2}(\frac{\pi^2}{4} - 2)\big) = \frac{1}{4}\big(2 - \frac{8}{\pi^2}\big)$.

From horizontal to Euclidean distance (outline). It remains to show that $w\,\mathbb{E}\,\delta$ and $w\,\mathbb{E}h$ have the same limit. Clearly $\delta \le h$. For the other direction fix $\eta \in (0,1)$ and suppose $|y_0| \le 1 - \eta$. Each monotone stroke of the curve is the graph of x as a function of the height y, with derivative $1/(w\pi\sqrt{1-y^2})$. On the window of heights within $2/w$ of y_0 this derivative is at most $\kappa = 1/\big(w\pi\sqrt{1-(1-\eta+2/w)^2}\big)$ once $2/w < \eta$, and $\kappa \to 0$ as $w \to \infty$. Points of the curve outside the window are further than $2/w \ge h$ from (x_0, y_0). Inside the window, Theorem 1 applies to each stroke with the roles of x and y exchanged, and gives $\delta \ge h/\sqrt{1+\kappa^2}$. The points with $|y_0| > 1 - \eta$ have probability η and $\delta \le h \le 2/w$, so they change $w\,\mathbb{E}\,\delta$ and $w\,\mathbb{E}h$ by at most 2η each. Letting $w \to \infty$ and then $\eta \to 0$ gives the claim. We have not written out the measurability and edge details of this last step, and for that reason label the result as leading-order; the numerical check is in Section 3.3.

A.5 The band-edge floor

We prove Proposition 9.

Lower bound. The curve lies in the band $|y| \le p$. A point (x_0, y_0) is therefore at distance at least $\max(0, |y_0| - p)$ from it. For y_0 uniform on $[-1,1]$, $\mathbb{E}\max(0,|y_0|-p) = \int_p^1 (t-p)\,dt = (1-p)^2/2$.

Upper bound. We use the following fact: if $\Lambda \subset \mathbb{R}$ is a lattice of period $2/w$ with $w \ge 1$, every $x_0 \in [-1,1]$ has a point of $\Lambda \cap [-1,1]$ within $2/w$. Indeed, let $x_L \le x_0 < x_R = x_L + 2/w$ be the neighbours of x_0 in Λ. If $x_L \ge -1$ take x_L. Otherwise $x_R < -1 + 2/w \le 1$ and $x_R > x_0 \ge -1$, so take x_R.

If $|y_0| \le p$, the crossings of the curve at height y_0 contain a lattice of period $2/w$, so there is a point of the curve at height y_0 within $2/w$ of x_0, and the distance is at most $2/w$. If $y_0 > p$, the maxima of the curve form a lattice of period $2/w$ at height p, so the distance is at most $\sqrt{(y_0-p)^2 + (2/w)^2} \le (y_0 - p) + 2/w$. The case $y_0 < -p$ is the same with the minima. In every case the distance is at most $\max(0, |y_0| - p) + 2/w$, for every phase. Taking expectations gives the upper bound. □

B What went wrong in the first version

The first scripts, written in August 2025, appeared to support the conjecture, and a draft paper reported that support. Three of their measurements were not what they were described as. The repository keeps the scripts unchanged, with a warning, and a script that reproduces each defect.

1. *The "distance" was a neuron count.* The core metric compared the label with the number of active neurons of the network, which is about half the width. It measured 4.9 at width 10 and 80.1 at width 160. It grows linearly with width, the opposite of the conjecture, and was captioned as support for it. Its apparent independence of the input dimension was the absence of any dependence on the data.
2. *The scaling law was the output fading.* A power law with exponent about 0.42 was reported for $\|\mathrm{net}(x) - x\|$, the distance from the output of a network to its own input, on a $2 \to w \to w \to 2$ ReLU stack with Xavier initialisation [Glorot and Bengio, 2010]. In that setup the output shrinks towards zero as the width grows, so the metric converges to $\|x\|$, the score of a network that returns nothing: 1.3186 at width 2 and 1.2325 at width 128, against 1.2299 for the zero function.
3. *The one-dimensional case measured against a diagonal.* One coordinate of each random point was used as both x and y, which placed every point on the line $y = x$.

The original derivation also estimated the mean distance as a power of the uncovered fraction of the region and drew a conclusion about diminishing returns that the formula does not support; Theorem 5 replaces that estimate.

Two later corrections were numerical and concern the experiments reported here.

1. *Fixed sampling grid (corrected 19 September 2026).* Curves were first sampled on a fixed grid of 4000 points. At $w = 64$ the steepest slope of the sine is about 164, so consecutive samples were about 0.08 apart vertically, more than the distances being measured. The nearest-point distance was overestimated from $w \approx 32$ upwards (0.0279 reported at $w = 64$, against 0.0207), which produced a spurious plateau. The grid is now sized for each frequency.
2. *Wrong stroke (corrected 11 November 2026).* The replacement refined the distance only on the two segments next to the nearest sample. Once the gap $1/w$ between neighbouring strokes is below the sample spacing 0.002, the nearest sample can lie on the wrong stroke. In the control of Section 3.3 this overestimated the mean distance by 11% at $w = 1000$ (0.000324 against 0.000292). The routine now projects onto every segment that can contain the nearest point, is exact for the polyline, and agrees with a brute-force reference to better than 10^{-6} from $w = 0.25$ to $w = 1000$ at both amplitudes. Tables 1 and 2 did not change in the digits shown. The tail fits of Section 3.3 moved slightly: before the correction they read $\beta = 0.978$, $c = 0.236$, a predicted knee of 55.7, and $0.285\, w^{-0.989}$ for the control.

C Experimental details

All scripts are in the `scripts/` folder of the repository and use fixed random seeds.

- `frequency_isolation.py`: Tables 1 and 2. 3000 trials of 30 points; 100 stratified phases per frequency; about ten seconds.
- `saturation_scaling.py`: Table 3 and the fits of Section 3.3. $30\,000$ points per n, 16 phases per frequency, 25 frequencies logarithmically spaced in $[1, 1000]$, 300 bootstrap resamples; the control uses $n = 100$. Several minutes.
- `initial_mse.py`: Table 4.
- `random_relu_length.py`: Table 5.
- `curves.py`: the sine geometry and the exact point-to-polyline distance shared by the first two scripts.
- `audit_legacy_metrics.py`: reproduces the three defects of Appendix B.
- `readme_figures.py`: draws Figures 1 to 4 from the same functions.

The numbers in Tables 1 to 5 were reproduced by running these scripts for this paper. The slope and share-outside columns of Table 2, the bound columns of Table 1 and the check of Proposition 8 quoted in Section 3.3 were computed for this paper from the same functions. Over the 1600 networks that the last script pools across sizes, the correlation between arc length and nearest-point distance is -0.76.

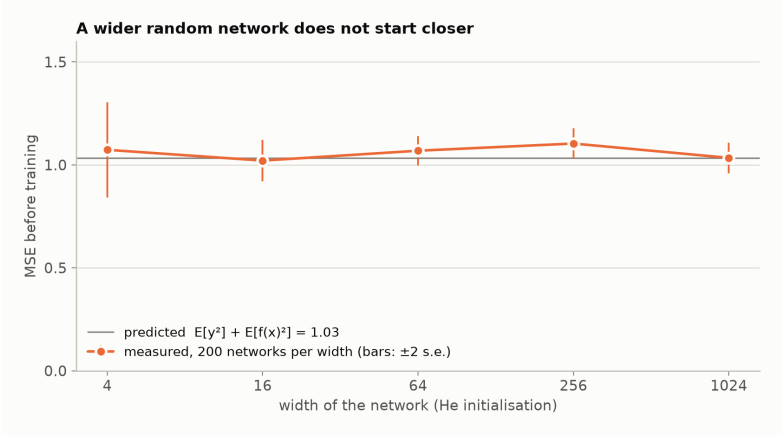


Figure 4: Initial squared error of random ReLU networks under He initialisation (Table 4), with the prediction of Corollary 4. Bars are ± 2 standard errors.

Table 5: Random ReLU networks with a one-dimensional input: all sizes ($\pm$ one standard error over 300 networks; no training). Setup and caveats as in Section 3.4. Reference curves measured by the same code: the diagonal $y = x$ has length 2.83 and distance 0.520; a sine with one period across the box, 4.61 and 0.351; a sine with 16 periods, 64.07 and 0.021.

width	hidden layers	parameters	arc length	turns	nearest-point
2	2	12	3.446 ± 0.040	0.47	0.4808 ± 0.0052
4	2	32	3.571 ± 0.037	1.47	0.4540 ± 0.0050
8	2	96	3.512 ± 0.034	2.42	0.4398 ± 0.0049
16	2	320	3.604 ± 0.035	3.16	0.4231 ± 0.0046
32	2	1152	3.571 ± 0.034	4.39	0.4269 ± 0.0046
64	2	4352	3.559 ± 0.035	5.89	0.4269 ± 0.0045
128	2	16896	3.636 ± 0.037	8.18	0.4179 ± 0.0046
256	2	66560	3.577 ± 0.036	11.54	0.4286 ± 0.0047
512	2	264192	3.599 ± 0.034	15.11	0.4257 ± 0.0046
64	1	192	3.417 ± 0.034	3.05	0.4428 ± 0.0047
64	2	4352	3.613 ± 0.037	5.90	0.4256 ± 0.0047
64	3	8512	3.761 ± 0.037	8.39	0.4116 ± 0.0048
64	4	12672	3.900 ± 0.039	10.85	0.3938 ± 0.0044
64	6	20992	4.058 ± 0.043	15.67	0.3882 ± 0.0044
64	8	29312	4.368 ± 0.049	20.80	0.3643 ± 0.0042
64	12	45952	4.865 ± 0.062	28.88	0.3431 ± 0.0040
64	16	62592	5.100 ± 0.075	36.46	0.3433 ± 0.0046